Species-Aware Review: Lol-Dependent Outer-Membrane Lipoprotein Trafficking in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Bacterial Lol-dependent OM lipoprotein trafficking (LolCDE → LolA → LolB) **Commissioned bucket:** KEGG `ppu02010` "ABC transporters" (207 candidate genes) **Date:** 2026-09-01

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1. Executive Summary

The Lol module is **fully satisfiable** in *P. putida* KT2440. Every expected step maps one-to-one to a high-confidence, single-copy ortholog (LolD, LolA, LolB are Swiss-Prot/HAMAP-reviewed; LolC, LolE are TrEMBL but confirmed by synteny and 40–46 % identity to *E. coli* — see §4):

Module step	Gene	Locus	UniProt	KO
LolCDE IM extraction/release (permease)	*lolC*	PP_2154	Q88KY5	K09808
LolCDE IM extraction/release (ATPase)	*lolD*	PP_2155	Q88KY4 (LOLD_PSEPK)	K09810
LolCDE IM extraction/release (permease)	*lolE*	PP_2156	Q88KY3	K09808
LolA periplasmic carriage	*lolA*	PP_4003	Q88FS9 (LOLA_PSEPK)	K03634
LolB OM delivery/incorporation	*lolB*	PP_0724	Q88PX4 (LOLB_PSEPK)	K02494

Two curation-critical points:

1. **The provided candidate list does not represent this module.** None of the five Lol genes appear in the `kegg:ppu02010` candidate set. `lolA`/`lolB` are not ABC transporters, and `lolCDE`, although mechanistically an ABC transporter, is **not assigned to any KEGG**

pathway map (its KEGG `PATHWAY` field is empty for all three genes). The Lol system is a KEGG **BRITE/orthology** entity, not a `map02010` member. Curators must source this module by KO or GO:0044874, **not** by the ABC-transporter bucket.

2. **All steps should be marked `covered`.** There is no gap, and no lineage-specific replacement is needed — unlike some diderms that lack *lolB* and rely on a bifunctional LolA, KT2440 retains a genuine *lolB*.

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2. Target-Organism Pathway Definition

Included biochemistry (this module): ATP-dependent release of *mature, OM-destined* triacylated lipoproteins from the outer leaflet of the inner membrane by the LolCDE ABC complex; hand-off to the periplasmic carrier LolA; transfer across the periplasm; and receptor-mediated insertion into the inner leaflet of the OM by LolB.

Explicitly upstream / separate (keep out of this module): - Lipoprotein maturation and IM-retention decisions: Lgt (diacylglyceryl transferase), LspA/signal peptidase II, Lnt (N-acyl transferase), and the "+2 rule" Lol-avoidance signal (Zückert 2014, P 24780125). These are *prerequisites*, not Lol steps. - Sec/general secretory export of prolipoproteins. - Downstream OM assembly machines that *consume* Lol cargo: Bam (β-barrel assembly), Lpt (LPS export — note LptB PP_0953, LptF PP_0982, LptG PP_0983 are in the candidate list but belong to a **different** module), Lpo–PBP.

Neighboring overview maps to keep separate: KEGG `map02010` (ABC transporters) and `map03070` (secretion). The Lol system overlaps mechanistically with ABC transporters but should be curated as its own transport/localization module.

Alternate names / definitions: "Lipoprotein-releasing system" (KEGG/GenBank naming for LolCDE), "Lol pathway/machinery," "outer-membrane lipoprotein localization (Lol)."

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3. Expected Step Model vs. Target Taxon

Generic step	Player	Status in KT2440	Evidence
1. ATP-dependent IM extraction/release	LolCDE	**covered**	*lolC-lolD-lolE* syntenic operon PP_2154–2156; HAMAP-reviewed LolD; KO K09808/K09810
2. Periplasmic carriage	LolA	**covered**	*lolA* PP_4003, HAMAP MF_00240, single-copy, periplasmic
3. OM delivery/ incorporation	LolB	**covered**	*lolB* PP_0724, HAMAP MF_00233, OM lipid-anchor, single-copy

Step-relationships (LolCDE→LolA→LolB) are conserved and structurally characterized in the paradigm organisms (Kaplan et al. 2018/2022,

P 30012603

P 36037338

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4. Candidate Genes and Evidence

Because the `ppu02010` candidate list omits all Lol genes, the relevant "candidates" are the KO/ UniProt-derived orthologs:

- **PP_2154 *lolC* (Q88KY5, K09808):** IM permease, MacB/FtsX-like. Evidence: orthology + HAMAP-family GO annotation (GO:0044874). Caveat: MacB/FtsX permease family is large; BLAST alone can confuse LolC with MacB efflux permeases or TagS-like T6SS proteins. Synteny with *lolD/lolE* is the discriminating feature.
- **PP_2155 *lolD* (Q88KY4, LOLD_PSEPK, K09810):** ABC ATPase, EC 7.6.2.-, HAMAP MF_01708 (reviewed). **Highest confidence** of the operon. Caveat: EC 7.6.2.- and generic "ABC ATP-binding" annotations are broad; do not transfer LolD identity to other ABC ATPases on EC alone.
- **PP_2156 *lolE* (Q88KY3, K09808):** second IM permease; paralogous fold to LolC. Same MacB/ FtsX-family caveat.

- **PP_4003 *lolA* (Q88FS9, LOLA_PSEPK, K03634):** periplasmic carrier, HAMAP MF_00240 (reviewed). Single-copy — no paralog ambiguity (confirmed by family and text searches returning exactly one hit).
- **PP_0724 *lolB* (Q88PX4, LOLB_PSEPK, K02494):** OM lipoprotein receptor, HAMAP MF_00233 (reviewed). Single-copy. DUF1329 hits (PP_0766, PP_2043, PP_2810, PP_5590) are **not** LolB — different fold, spurious text matches.

Genomic context: *lolCDE* sits between PP_2153 (PilZ-domain protein) and PP_2157 (sensor histidine kinase); a clean three-gene operon. *lolA* and *lolB* are unlinked (as in most gamma-proteobacteria).

Quantitative orthology check (Needleman–Wunsch % identity, KT2440 vs *E. coli* K-12):

Protein	KT2440	*E. coli*	% identity
LolC	PP_2154 (Q88KY5)	P0ADC3	45.9
LolD	PP_2155 (Q88KY4)	P75957	61.2
LolE	PP_2156 (Q88KY3)	P75958	40.5
LolA	PP_4003 (Q88FS9)	P61316	39.7
LolB	PP_0724 (Q88PX4)	P61320	30.3

All identities are well above the ~20–25 % twilight zone; the gradient (LolD ATPase most conserved → LolB OM receptor least conserved) matches known Lol evolutionary rates. This confirms the two TrEMBL entries (PP_2154 LolC, PP_2156 LolE) are true Lol permeases, **not** MacB efflux paralogs (which score far lower vs *E. coli* LolC/E). Sequence features corroborate function: **LolB** N-terminus `MFLRHCITFTLIALLAGC…` carries a lipobox (+1 Cys after `LAGC`) — LolB is itself a lipoprotein and thus an *autologous* Lol client; **LolA** N-terminus `…VTAYA↓G…` is a cleavable Sec signal peptide with an AXA signal-peptidase-I site and no lipobox — a soluble periplasmic carrier.

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5. Gaps, Ambiguities, and Likely Over-Annotations

- **No true biological gap.** All three module steps covered by reviewed orthologs.

- **Paralog / over-propagation risk (LolCDE only) — concrete KT2440 loci:** LolC/LolE share the MacB/FtsX permease fold and LolD shares EC 7.6.2.- with several unrelated KT2440 ABC systems. Verified MacB/FtsX-family paralogs that must **not** be labeled Lol: PP_3734/PP_3735 and PP_0506/PP_0507 (MacB-type efflux), PP_2316/PP_2317 (*ybbA*; note PP_2317's annotation even text-matches "lipoprotein releasing"), *pvdT* PP_4210 (pyoverdine export, EC 7.6.2.-), and cell-division **FtsEX** (*ftsX* PP_5109 / *ftsE* PP_5110). *P. aeruginosa* additionally has a LolCDE-homologous **TagTS** ABC dedicated to type VI secretion (Casabona et al. 2013, P 22765374); KT2440 carries **three** T6SS clusters (HSI-I ~PP_2614–2627, HSI-II ~PP_3088–3106, HSI-III PP_4049/PP_4071–4081) but no clearly annotated dedicated TagTS. Anchor Lol identity to the syntenic **PP_2154–2156** operon + KO K09808/K09810, never to fold or EC.
- **Cargo vs. machinery:** the three T6SS OM lipoproteins **TssJ** (PP_2618, PP_3094, PP_4079) are Lol-dependent *clients*, not Lol components — curate them downstream of, not within, this module.
- **Broad-mapping caveats:** LolD's EC 7.6.2.- and the generic "ABC transporter, ATP-binding protein" descriptors in the candidate list (e.g., PP_0141, PP_1078) are not Lol-specific.
- **Candidate-bucket mismatch (metadata defect, not biology):** the `ppu02010` extraction is the wrong source for this module (Section 1, point 1).

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6. Module and GO-Curation Recommendations

- **Step 1 (LolCDE):** `covered` — PP_2154/2155/2156.
- **Step 2 (LolA):** `covered` — PP_4003.
- **Step 3 (LolB):** `covered` — PP_0724.
- **Module status:** satisfiable/covered; **not** `gap` or `not_expected_in_target_taxon`.
- **`module_needs_revision` (metadata, not biology):** the candidate-gene provisioning rule should pull Lol genes by KO (K09808, K09810, K03634, K02494) or GO:0044874, replacing the `kegg:ppu02010` bucket, which returns 207 ABC-transporter genes and zero Lol genes. **Verified:** genome-wide KEGG KO→gene links return *exactly* the five Lol genes and nothing else — K09808→PP_2154+PP_2156, K09810→PP_2155, K03634→PP_4003, K02494→PP_0724 — with the MacB/FtsX efflux paralogs excluded (they carry different KOs). KO-keyed sourcing is therefore clean and single-copy; no manual paralog filtering is needed.

- **GO:** existing GO:0044874 (lipoprotein localization to OM), GO:0042953 (lipoprotein transport), GO:0030288 (OM-bounded periplasm) are already applied and adequate; **no new GO term requests needed**.
- **Boundary correction:** ensure Lpt genes present in the candidate list (PP_0953 *lptB*, PP_0982, PP_0983) are curated to the LPS-export module, not conflated with Lol.

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7. Genes to Promote to Full `fetch-gene` Review

Priority order: 1. **PP_2154 *lolC*** and **PP_2156 *lolE*** — confirm they are the housekeeping Lol permeases (not MacB/TagS paralogs) via synteny and reciprocal-best-hit; these are the least "reviewed" of the set (TrEMBL, not Swiss-Prot). 2. **PP_2155 *lolD*** — verify operon co-transcription; already Swiss-Prot, lower risk. 3. **PP_0724 *lolB*** and **PP_4003 *lolA*** — Swiss-Prot reviewed and single-copy; promote mainly to attach direct citations and confirm essentiality expectation.

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8. Key References

- Zückert WR. Secretion of bacterial lipoproteins… *BBA* 2014.

  P 24780125

  — defines Lol scope and upstream maturation boundary.
- Grabowicz M. Lipoproteins and their trafficking to the OM. 2019.

  P 30900542

  — LolCDE/LolA/LolB paradigm.
- Kaplan E, et al. LolA–LolC structure. *PNAS* 2018.

  P 30012603

  ; Structural basis of recognition by LolA. 2022.

  P 36037338

  — mechanism of hand-off.

- Smith AN, et al. Evolution of lipoprotein trafficking; bifunctional LolA. 2023.

  P 36716372

  — lineage variation, *lolB* loss in some diderms (KT2440 retains *lolB*).
- Hoang HH, et al. OM targeting in *P. aeruginosa* depends on Bam/Lol. 2011.

  P 22147293

  — genus-level functional evidence; LolB depletion affects all lipoproteins.
- Casabona MG, et al. LolCDE-homologous TagTS activates T6SS in *P. aeruginosa*. 2013.

  P 22765374

  — paralog/over-annotation caveat.
- Molina-Henares MA, et al. Conditionally essential genes of *P. putida* KT2440. 2010.

  P 20158506

  — KT2440 functional-genomics context.

Evidence-confidence summary

- **Direct for target strain:** genome/proteome annotation (UniProt UP000000556, KEGG ppu) — high confidence, but computational/homology-based; no published KT2440-specific Lol knockout phenotype located.
- **Strong genus transfer:** *P. aeruginosa* functional data (P 22147293) P 22765374 — same genus, syntenic single-copy orthologs.
- **Paradigm mechanism:** *E. coli* structural/biochemical work — mechanism, not strain-specific claims.

Open questions: (i) experimental confirmation that *lolCDE* is one operon in KT2440; (ii) direct essentiality of *lolA*/*lolB* in KT2440 (expected essential, not in the minimal-medium conditional screen of P 20158506); (iii) whether any KT2440 MacB/FtsX paralog carries a mis-propagated Lol label.